SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO2 – Manipulation(BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	20% of CIA	**Total Marks:**	50 (5 Questions × 10 Marks)
CO2 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)*		
Student Name:	Krishna Tejesh Reddy	**Reg. No.:**	192411200
Section / Batch:	S10E-C	**Date:**	1-08-2026

Code of Conduct (192411200).

I Krishna Tejesh Reddy *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10

		GRAND TOTAL:	50 Marks

Annexure A – ANALYTICAL PROBLEM SOLVING

1.Grey Level Transformation

A grayscale image is enhanced using a negative transformation.

(a) Explain the purpose of grey level transformation in image enhancement

(b) Apply the negative transformation:

$$s = 255 - r$$

to the following pixel values:

$$r = 50, 100, 150, 200$$

Compute the transformed pixel values and interpret the effect on the image.

2.Histogram Equalization

A 4-bit grayscale image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

(a) Explain why histogram equalization is used for image enhancement.

(b) Compute the cumulative distribution function (CDF).

(c) Determine the equalized intensity values and interpret the improvement in image contrast.

3.Gaussian Smoothing

A Gaussian filter is applied to reduce image noise.

(a) Explain why Gaussian filtering is preferred over averaging filtering.

(b) Apply the Gaussian kernel

$$\frac{1}{16}[1\ 2\ 1\ 2\ 4\ 2\ 1\ 2\ 1]$$

to the image region

$$[10\ 20\ 10\ 20\ 40\ 20\ 10\ 20\ 10]$$

Compute the new center pixel value and interpret the smoothing effect.

4.Edge Detection Using Sobel Operator

An image is processed using the horizontal Sobel operator.

(a) Explain the concept and purpose of edge detection.

(b) Apply the horizontal Sobel mask

$$[-1\ -2\ -1\ 0\ 0\ 0\ 1\ 2\ 1]$$

to the image

$$[10\ 10\ 10\ 20\ 20\ 20\ 30\ 30\ 30]$$

Compute the gradient at the center and interpret the result.

5.Contrast Stretching

A grayscale image has intensity values in the range 50–150.

(a) Explain the need for contrast stretching in image enhancement.

(b) Using linear contrast stretching to map the range [50,150] to [0,255], compute the transformed values for

$$r = 50, 100, 150$$

Show the calculations and interpret the improvement in image contrast.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method		
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process		
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations		
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

Name:- Y. Krishna Tejesh Reddy
Reg No:- 192111200
Course:- ITA05

CO2 - Assessment Tool - 1

Analytical Problem Solving

1. a. Grey level transformation enhances the brightness and adjusts it due to the the intensities in the image changes.

30/50

b. negitive transformation.

$$\boxed{S = 255 - r}$$

given r values 50, 100, 150, 200.

5

$S_1 = 255 - 50 = 205$

$S_2 = 255 - 100 = 155$

$S_3 = 255 - 150 = 105$

$S_4 = 255 - 200 = 55$

2. a. Histogram equalization is used to spread pixel intensities more evenly of an image.

b. Total No. of pixels is = 26

$\Rightarrow 0 \longrightarrow 4$ CDF 4

$\Rightarrow 1 \longrightarrow 6$ CDF 10

$\Rightarrow 2 \longrightarrow 10$ CDF 20

$\Rightarrow 3 \longrightarrow 6$ CDF 26

6

c. $I = \frac{CDF}{total}(L-1)$ I = Intensity, L = 16 (4-bit image)

$0 = \frac{4}{26}(16-1)$ $\Rightarrow 0 = \frac{2}{13}(15) \approx 2.1 \approx 2$

$1 = \frac{6}{26}(15) \approx 6$ | $\Rightarrow 2 \rightarrow \frac{20}{26}(15) \approx 12$ | $\Rightarrow 3 \Rightarrow \frac{26}{26}(15) = 15$

3. a. Gaussian filtering is preffered over averaging filtering beacause it gives more weight to the neighboring pixels and reduces the noise in the image.

b. given

$\frac{1}{16}$ [1 2 1 2 4 2 1 2 1] to [10 20 10 20 40 20 10 20 10]

$$= \frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

weighted sum at centre.

$\Rightarrow (1\cdot 10 + 2\cdot 20 + 1\cdot 10 + 2\cdot 20 + 4\cdot 40 + 2\cdot 20 + 1\cdot 10 + 2\cdot 20 + 1\cdot 10)/16$

$\Rightarrow (10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10)/16 = 22.5$

4. a. Purpose

Edge detection highlights boundaries where intensity changes sharply, useful for object recognition.

b. Horizontal Sobel mask:-

[-1, -2, -1 ; 0, 0, 0, 1, 2, 1]

Image region $\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$

Gradient at centre.

$(-1\cdot 10 + -2\cdot 10 + -1\cdot 10) + (1\cdot 30 + 2\cdot 30 + 1\cdot 30)$

$= (-10 - 20 - 10) + (30 + 60 + 30)$

$= -40 + 120 = 80$

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5 a. when intensity range is narrow (50 - 150) image looks dull. streatching maps it to full range (0-255)

b. given

$r = 50, 100, 150.$

formula.

$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

6

1) $r = 50 \longrightarrow s = 0$

2) $r = 100 \longrightarrow s = \frac{50}{100} \times 255 = 127.5 \approx 128$

3) $r = 150 \longrightarrow s = 255$